ROTATION: KINDERGARTEN

Chapter 6 — Kindergarten Rotation

Health assessment and screening of the preschool child (3-6 years): site role and readiness concepts, the preschool history and physical, vision and hearing screening with referral criteria, developmental and school-readiness screening, play and socialization, common problems, and health-education activities.

Age range: 3 – 6 years

6.1 Site Profile — Role of the Kindergarten

Preschool should be used primarily as an opportunity to **foster the child's social skills** and accustom the child to the group environment. Selecting a preschool considers accreditation, teachers' qualifications, other parents' recommendations, and the structure of the daily schedule.

- **What the nurse brings:** the preschooler "shows off" dramatically and enjoys imaginative, dramatic play — assessment and teaching are built around play.
- **Readiness principle:** there are **no absolute indicators** for school readiness; the child's social maturation, especially attention span, is as important as academic readiness.
- **Readiness is influenced by:** social, emotional, and physical development; health status; ability and desire to learn; life experiences; family environment; and parental support.
- **Risk factors to screen for:** insecure attachment in the early years, maternal depression, parental substance abuse, and low socioeconomic status — screen and refer when present.
- **By kindergarten entry** the child should have developed cooperation, sharing (of things and feelings), kindness, generosity, affection display, conversation, expression of feelings, helping others, and making friends.

6.2 Preschool Health Assessment

The well-child clinic model (Ch. 4) applies: history, measurements, screening, then guidance. The preschooler can now cooperate with instructions, so the examination is more efficient.

Step	Content
History	Interval history — feeding/elimination/sleep/behavior, past illness, allergies (incl. food), medications, immunization record; school/daycare attendance and concerns
Measurements	Length/height, weight, head circumference; now standing height usually (length board until 24 mo; tape not used); BMI from ~3 yr; blood pressure annually
Physical exam	Head-to-toe (Ch. 3/Ch. 4 technique); eyes — ocular alignment and acuity screening (6.3); ears — hearing screening (6.3); skin; teeth — caries check; abdomen, genitalia, extremities; developmental observation throughout
Screening	Vision (6.3), hearing (6.3), developmental screen / Denver II (6.4 role; technique in Ch. 2), and psychosocial risk factors (list above)
Immunization review	Check the record against the current printed schedule (see Ch. 4 §4.5); arrange catch-up; document
Guidance	Nutrition values, dental care, safety (6.6-6.7); reading and imaginative play promotion (Ch. 4 §4.4 preschool)

6.3 Vision & Hearing Screening

VISION

- **Indication:** the USPSTF recommends vision screening for the presence of amblyopia and its risk factors for **all children 3-5 years**.
- **Alignment tests:** corneal light reflex (Hirschberg) from about 40.5 cm (16 in) — light symmetrical in normal eyes, off-center in misalignment; epicanthal folds can mimic misalignment (pseudostrabismus, common in Asian children). Cover test at near (33 cm / 13 in) and far (6 m / 20 ft); alternate cover test.
- **Strabismus window:** if misalignment is not detected and corrected by ages **4-6 years**, blindness from disuse (amblyopia) may result.
- **Acuity:** the tumbling E and HOTV are excellent tests for **preschool-age children**; other options: Snellen letters/numbers, picture test, Allen figures, LEA symbols.
- **Referral criterion:** fewer than 4 of 6 correct on the 20-foot (6-m) line.
- **Peripheral vision** (cooperative child): normal ~50° upward, 70° downward, 60° nasalward, 90° temporally.

HEARING

- **Recommendation:** the AAP recommends **pure-tone audiometry** screening at 0.5, 1, 2, and 4 kHz; a child fails if the tones cannot be heard at **20 dB HL**.
- **Age-appropriate methods:** behavioral audiometry for toddlers; play audiometry for the toddler; pure-tone audiometry for children and adolescents; tympanometry (middle-ear function) at all ages; EOAE for cochlear (outer hair cell) response.
- **Suspicion:** maintain a high index of suspicion for children with conditions associated with hearing loss, whose parents report concern, or whose behavior suggests auditory impairment.

Figure 6.1 shows the three referral triggers of the preschool screening program: an acuity below the 20-foot-line standard, any constant misalignment (especially within the amblyopia window before ages 4-6), or failure of the pure-tone hearing screen.

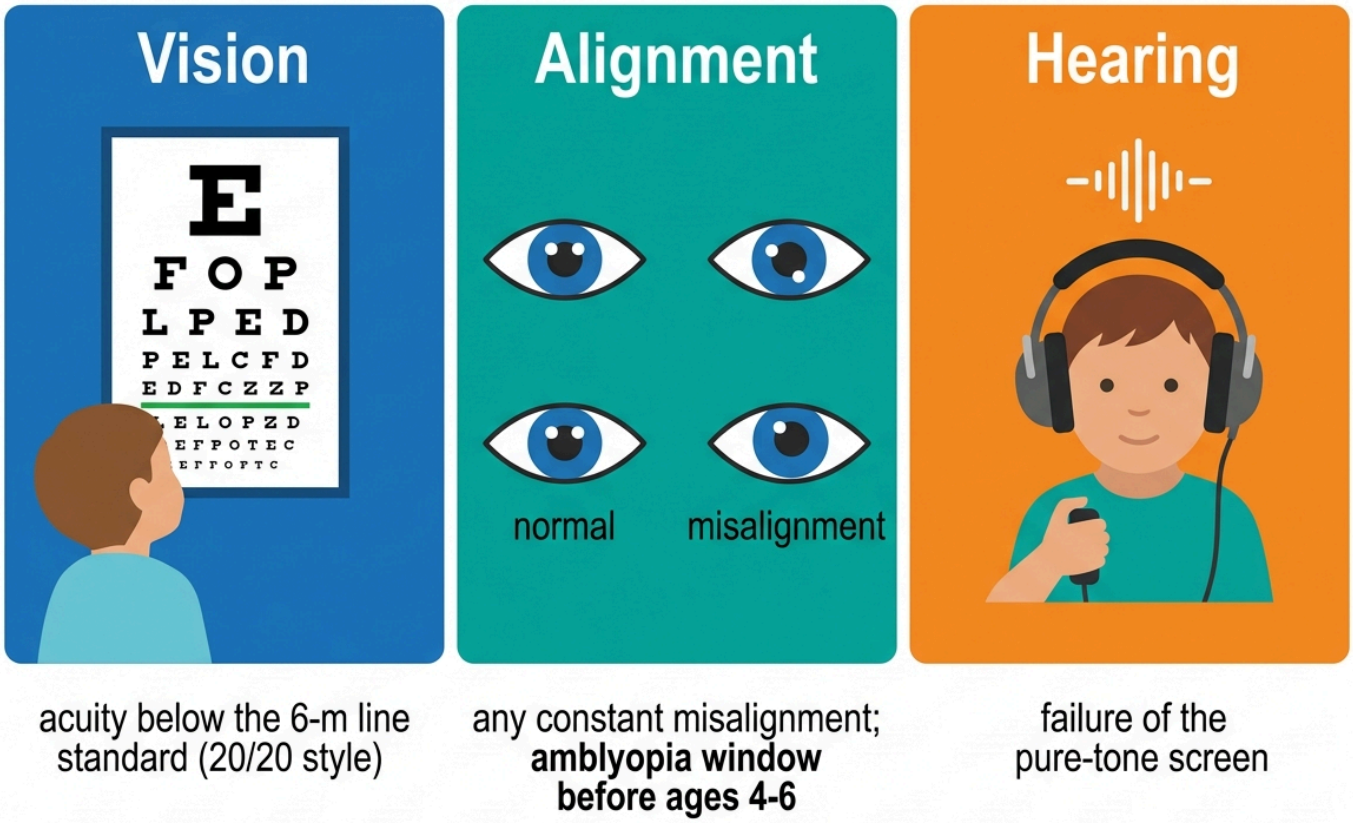


Figure 6.1 Referral triggers of the preschool vision and hearing screening program.

6.4 Developmental & School-Readiness Screening

Use the Denver II (technique in Ch. 2) supplemented by the milestone ages below. These are the 3-5-year motor and language landmarks to observe — a deficit against them draws the eye for formal testing.

Age	Gross motor	Fine motor / self-care
3 years	Rides a tricycle; walks on tiptoe; balances on one foot a few seconds; broad jumps	Copies a circle; imitates a cross and vertical/horizontal lines; holds the crayon with fingers (not fist); names what he/she draws

4 years	Throws a ball overhand; kicks forward; catches a bounced ball; hops on one foot; stands on one foot up to 5 s; alternates feet on steps	Uses scissors successfully; copies capital letters; draws circles and squares; traces a cross or diamond; draws a person with 2-4 body parts; laces shoes
5 years	Stands on one foot 10 s or longer; swings and climbs well; may skip; somersaults; may begin to skate and swim	Prints some letters; draws a person with body plus at least six parts; dresses/undresses without assistance; can learn to tie laces; uses fork/spoon/knife (supervised); copies triangle and other geometric patterns; mostly self-toileting

Age	Language / communication
4 years	Complete sentences with adult-like grammar; story easy to follow; 75% of speech understood outside the family; "who/how/how many" questions; understands same/different; names familiar animals and common objects; knows at least one color; engages in make-believe; follows a three-part command; counts a few numbers; vocabulary ~1,500 words
5 years	Most speech understood outside the family; explains how an item is used; long detailed conversations; talks about past/future/imaginary events; answers "why/when"; counts to 10; recalls part of a story; speech intelligible even with articulation errors; generally grammatical; vocabulary ~2,100 words; says name and address
Bilingual child	By 4 years the mixing of languages during the toddler years ceases — each language is used as a separate system

SPEECH REFERRAL NOTE

Stuttering with no improvement over more than 6 months, or speech unintelligible to outside listeners beyond the expected age, prompts evaluation (see the red-flag principle in Ch. 4 §4.6). Referral thresholds for the district screening policy follow the official screening program.

6.5 Play, Behavior & Socialization

- **Associative play:** preschoolers especially enjoy **associative play** — group play in similar or identical activities; the parallel play of the toddler (alongside, not with others) gives way to group activity.
- **Imagination:** activity is imitative, imaginative, and dramatic; dress-up is central; imaginary friends appear 2½-3 years and last until school entry; most fears resolve by 5-6 years.

- **Sharing & friendships:** by kindergarten entry the child cooperates, shares (things and feelings), shows kindness/generosity/affection, holds conversation, expresses feelings, helps others, and makes friends.
- **Separation:** during the preschool period the separation-individuation phase matures; preschoolers tolerate brief separations with little or no protest — the nurse normalizes this for worried parents.
- **Discipline:** the preschooler responds to consistent limit-setting with simple, concrete explanations — words are taken literally (magical thinking), so the parent says "that was a bad thing to do," not "you are bad".

6.6 Common Problems in the Kindergarten

Problem	Guidance
Infections	Children commonly develop infections when entering day care or preschool because of new exposure; symptom/exclusion policy per the facility (see Ch. 5)
Dental caries	Caries are present in ~60% of children by age 5; prevention = daily brushing and flossing, pea-sized fluoride toothpaste without excess (fluorosis), parent performs flossing, avoid cariogenic foods — rinse with water after sugary foods
Accidental ingestion & injury	Accidental injury is the leading cause of death in children 1-14 years; preschoolers can absorb concrete safety teaching but may not transfer it — close supervision continues; accidental ingestion risk is high — hazardous substances stored out of reach, never coax swallowing
Nutrition	Preschool nutrition values in Ch. 4 §4.4 (≈90 kcal/kg ≈ 1800 cal/day; fluids ~100 ml/kg/day; protein 13-19 g/day; fiber = age + 5; saturated fat <10%; calcium 700 mg 1-3 yr / 1000 mg 4-8 yr; juice cap 4-6 oz); obesity in young children has increased dramatically
Screening findings	Refer positive vision/hearing (6.3) and developmental findings (6.4); document and communicate with the family (Ch. 9)

6.7 Health Education Activities

- **Hand hygiene:** standard infection-control hand washing before and after every meal, diapering, and child contact.

- **Dental hygiene program:** demonstrate pea-sized toothpaste brushing; parents (not the child) floss; schedule dental examinations per policy.
- **Safety lessons:** an adult accompanies at street crossings; a bicycle helmet is worn on any ride; the child never approaches pools or bodies of water alone; medications and cleaning products are never touched.
- **Literacy:** daily reading/storytelling with question-and-answer supports language and school readiness.
- **Group readiness practice:** turn-taking, sharing, following three-part commands (a skill expected at 4 years).

Chapter 6 Learning Check

Self-assessment questions (answer in your notebook, then open)

End of Chapter 6.